

VALLABH SATHE

Kolhapur

| +91 8600889383

| sathe.vallabh12@gmail.com

| [LinkedIn](#)

PROFILE SUMMARY

Computer Science graduate and aspiring Full Stack Developer with hands-on experience in the MERN stack (MongoDB, Express.js, React, Node.js) and a growing focus on AI/ML. Skilled in building scalable web applications, RESTful APIs, real-time systems using [Socket.IO](#), and AI-powered features like recommendations and automation. Proficient in Python and machine learning fundamentals, with exposure to model training, deployment, and full-stack integration. Passionate about solving real-world problems and learning emerging technologies.

WORK EXPERIENCE

Full Stack Developer Intern

AlgoAnalytics Technologies Pvt. Ltd.

Jun 2025 — Present

- Designed and developed a fully responsive product showcase platform using React.js and Tailwind CSS, increasing user engagement by 40% 
- Architected and implemented secure authentication and user-management systems leveraging AWS Lambda, API Gateway, and DynamoDB in a scalable serverless environment.
- Developed a production-ready fire and smoke detection system using an end-to-end computer vision pipeline, integrated with Kafka-based streaming for real-time, low-latency processing.
- Solved the problem of occluded road sign detection by training YOLO-based object detection models on augmented datasets containing partially visible signs and optimizing model parameters to improve recognition under obstruction conditions.

EDUCATION

Sanjay Ghodawat University, Kolhapur

Bachelor of Technology in Computer Science and Engineering

Mar-2026

CGPA: 8.45

SKILLS

Programming Languages: Python, Java (Data Structures & Algorithms), C++

Cloud & DevOps: Docker, Git, GitHub, AWS, Lambda, API Gateway

Machine Learning & AI: NLP, LangChain, LangGraph, RAG, CrewAI, YOLO, scikit-learn, Model Training, Feature Engineering, Evaluation Metrics, MCP

Web Development: React.js, Node.js, Express.js, Next.js, Tailwind CSS, MongoDB, MySQL

PROJECTS

Student Course Recommendation System

Oct 2025 — Jan 2026

- Built a cold-start recommendation engine using a RAG architecture to retrieve and rank learning resources from GitHub, YouTube, and other platforms. Combined collaborative filtering with LLM-based suggestions to improve personalization for new users.
- Designed scalable pipelines for content ingestion, embedding generation, vector indexing, semantic search, and relevance scoring to deliver context-aware recommendations.
- Developed backend services in Python for model training, ranking logic, and real-time recommendation APIs with optimized performance.
- Containerized services using Docker and deployed on AWS, enabling scalable, production-ready recommendation and authentication systems.

Content Moderation & Community Safety Platform

Jul 2023 — Aug 2025

- Built an enterprise-grade AI content moderation platform using Multi-Agent Architecture (LangGraph, LangChain, Google Gemini) to automate analysis of user-generated content with explainable decision workflows.
- Designed and implemented a ReAct (Reason-Act-Observe) decision loop with 6 specialized AI agents (Toxicity, Policy, Reputation, Analysis, Decision, Enforcement) to synthesize moderation outcomes using consensus scoring.
- Engineered a Human-in-the-Loop (HITL) review system that auto-escalates low-confidence or high-severity cases into a priority moderation queue, enabling human override and continuous AI learning.
- Developed a full-stack moderation dashboard using React and Material-UI with role-based access control across 6 user roles, real-time review workflows, and explainable AI reasoning interfaces.

CERTIFICATIONS

Data Analytics with Python
Deep Learning - IIT Ropar
Machine Learning
React And JavaScript (Meta)

